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2-11

解：

1） 理想气体状态方程

$$pV_m = RT$$

$$V_m = \frac{RT}{p} = \frac{8.314 \times 473}{1.0 \times 10^6} = 3.93 \times 10^{-3} m^3 \cdot mol^{-1}$$

$$Z = \frac{pV_m}{RT} = 1$$

2）截项 virial 方程

$$V_m = \frac{RT}{p} + B = \frac{8.314 \times 473}{1.0 \times 10^6} + (-195 \times 10^{-6}) = 3.74 \times 10^{-3} m^3 \cdot mol^{-1}$$

$$Z = \frac{pV_m}{RT} = \frac{1.0 \times 10^6 \times 3.74 \times 10^{-3}}{8.314 \times 473} = 0.950$$

3）RK 方程

查表，水的 $T_c = 647.3K \qquad p_c = 22.05MPa$

则 $T_r = \frac{473}{647.3} = 0.7307 \qquad p_r = \frac{1.0}{22.05} = 0.0454$

则

$$q = \frac{\Psi}{\Omega} T_r^{-3/2} = \frac{0.42748}{0.08664} \times 0.7307^{-3/2} = 7.8993$$

$$\beta = \Omega \frac{p_r}{T_r} = \frac{0.08664 \times 0.0454}{0.7307} = 0.005383$$

对于水蒸汽，

$$Z = 1 + \beta - q\beta \frac{Z - \beta}{Z(Z + \beta)}$$

利用 $Z = 1$ 的初值进行迭代，收敛结果为 $Z = 0.962$

$$V_m = \frac{ZRT}{p} = \frac{0.962 \times 8.314 \times 473}{1.0 \times 10^6} = 3.78 \times 10^{-3} m^3 \cdot mol^{-1}$$

4)L-K 方程

$$Z = Z^0 + \omega Z^1$$

查表 $Z^0 = 0.9624 \qquad Z^1 = -0.03661 \qquad \omega = 0.344$

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$$Z=0.950 \qquad V_m=\frac{ZRT}{p}=3.74\times10^{-3}\cdot mol^{-1}$$

查水蒸气表，473K，1.0MPa 下水蒸汽（过热蒸汽）的摩尔体积为$3.71\times10^{-3}m^3\cdot mol^{-1}$

2-13

解：根据理想气体状态方程

$$p=\frac{nRT}{V}=1.638MPa$$

查表，$T_c=126.2K \qquad p_c=3.394MPa \qquad \omega=0.040$

则 $T_r=3.7480 \qquad p_r=0.4828$

查表 $Z^0=1.0048$

对照图 2/7-2，可知理想气体状态方程适用。

若用 L-K 方程，根据上面的条件计算出新的 p，进一步查表得到新的 Z 值，直至 p 收敛，结果 $p=1.647MPa$。

2-16

解：

查表，$T_c=425.2K \qquad p_c=3.799MPa \qquad \omega=0.193$

则 $T_r=1.0818 \qquad p_r=0.4001$

$$B^0=0.083-\frac{0.422}{T_r^{1.6}}=-0.28909$$

$$B^1=0.139-\frac{0.172}{T_r^{4.2}}=0.0154$$

$$Z=1+B^0\frac{p_r}{T_r}+\omega B^1\frac{p_r}{T_t}=0.8942$$

$$V_m=\frac{ZRT}{p}=2.250\times10^{-3}m^3\cdot mol^{-1}$$

若以 $B=-0.265m^3\cdot kmol^{-1}$ 计算，

$$V_m=\frac{RT}{p}+B=2.251\times10^{-3}m^3\cdot mol^{-1}$$

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相对偏差=0.06%

若 $p=1Mpa$，根据普遍的 virial 方法，$V_m=3.560m^3\cdot mol^{-1}$，根据实验值 $V_m=3.560m^3\cdot mol^{-1}$

2-19

解：

1）Antoine 方程

查表，$A=15.8333$　$B=2477.07$　$C=-39.94$

$$\ln p^s=A-\frac{B}{T+C}=15.8333-\frac{2477.07}{423.15-39.94}=9.3693$$

$$p^s=11722.89mmHg=1.542MPa$$

2）Pitzer 关联方程

查表，正戊烷的 $T_c=469.6K$　$p_c=3.374MPa$　$\omega=0.251$

则　$T_r=\frac{423.15}{469.6}=0.9011$

$$f^{(0)}(T_r)=5.92714-\frac{6.09648}{T_r}-1.28862\ln T_r+0.169347T_r^6=-0.6137$$

$$f^{(1)}(T_r)=15.2518-\frac{15.6875}{T_r}-13.4721\ln T_r+0.43577T_r^6=-0.5213$$

则　$\ln p_r^s=f^{(0)}(T_r)+\omega f^{(1)}(T_r)=-0.7445$

$$p_r^s=0.4749$$

$$p^s=0.4749\times3.374=1.602Mpa$$

2-18

解：查表，$T_c=425.2K$　$p_c=3.799MPa$　$\omega=0.193$　$V_c=255cm^3\cdot mol^{-1}$　$Z_c=0.274$

则 $T_r=\frac{T}{T_c}=0.9246$　$p_r=\frac{p}{p_c}=0.5791$

Rackett 方程

$$V=V_cZ_c^{(1-T_r)^{0.2857}}=137.37\times10^{-3}m^3\cdot kmol^{-1}$$

RK 方程

$$q=\frac{\Psi}{\Omega}T_r^{-3/2}=\frac{0.42748}{0.08664}\times0.7307^{-3/2}=5.5494$$

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$$\beta=\Omega\frac{p_r}{T_r}=\frac{0.08664\times0.0454}{0.7307}=0.05426$$

$$Z_{k+1}=\beta+Z_k(Z_k+\beta)(\frac{1+\beta-Z_k}{q\beta})$$

$Z_0=0$ 迭代，得 $Z=0.1149$

$$V=\frac{ZRT}{p}=170.68\times10^{-3}m^3\cdot kmol^{-1}$$

2-20

解：

Virial 方程

$$B=y_1^2B_{11}+y_2^2B_{22}+2y_1y_2B_{12}=-132.58cm^3\cdot mol^{-1}$$

$$V=\frac{RT}{p}+B=0.415m^3\cdot kmol^{-1}$$

RK 方程

查表 N_2 $T_c=126.2K$ $p_c=3.394MPa$

C_4H_{10} $T_c=425.2$ $p_c=3.799MPa$

对于 N_2，

$$T_r=3.6529$$

$$a=0.42748\frac{T_r^{-1/2}R^2T_c^2}{p_c}=0.07255Pa\cdot m^6\cdot K^{0.5}\cdot mol^{-2}$$

$$b=0.08664\frac{RT_c}{p_c}=26.7840cm^3\cdot mol^{-1}$$

对于 C_4H_{10}，

$$T_r=1.0842$$

$$a=0.42748\frac{T_r^{-1/2}R^2T_c^2}{p_c}=1.3505Pa\cdot m^6\cdot K^{0.5}\cdot mol^{-2}$$

$$b=0.08664\ \frac{RT_c}{p_c}=80.6218\ cm^3\cdot mol^{-1}$$

则

$$a=y_1^2a_1+y_2^2a_2+2y_1y_2\sqrt{a_1a_2}=0.7997Pa\cdot m^6\cdot K^{0.5}\cdot mol^{-2}$$

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$$b = y_1 b_1 + y_2 b_2 = 64.4705 cm^3 \cdot mol^{-1}$$

$$\beta = \frac{bp}{RT} = 0.1177$$

$$q = \frac{a(T)}{bRT} = 3.2365$$

$$Z = 1 + \beta - q\beta \frac{Z - \beta}{Z(Z + \beta)}$$

初值 1 迭代， $Z = 0.7463$

$$V = \frac{ZRT}{p} = 0.409 m^3 \cdot kmol^{-1}$$

3-22

解：

查表，水的 $T_c = 647.3K$ $p_c = 22.05MPa$ $\omega = 0.344$

则 $T_r = 0.7309$ $p_r = 0.0635$

$$B^0 = 0.083 - \frac{0.422}{T_r^{1.6}} = -0.61376$$

$$B^1 = 0.139 - \frac{0.172}{T_r^{4.2}} = -0.50247$$

$$B = \frac{RT_c}{p_c}(B^0 + \omega B^1) = -1.920 \times 10^{-4} m^3 \cdot mol^{-1}$$

故 $V^R = B = -1.920 \times 10^{-4} m^3 \cdot mol^{-1} = -0.0107 m^3 \cdot kg$

$$\frac{dB^0}{dT_r} = \frac{0.675}{T_r^{2.6}} = 1.5247 \qquad \frac{dB^1}{dT_r} = \frac{0.722}{T_r^{5.2}} = 3.6838$$

故 $H^R = -p_r RT[(\frac{dB^0}{dT_r} - \frac{B^0}{T_r}) + \omega(\frac{dB^1}{dT_r} - \frac{B^1}{T_r})] = -966.1 J \cdot mol^{-1}$

$$S^R = -p_r R(\frac{dB^0}{dT_r} + \omega \frac{dB^1}{dT_r}) = -1.4738 J \cdot mol^{-1} \cdot K^{-1}$$

查水蒸气表，得到实际气体的各项性质。理想气体的各项性质由三相点为零点开始计算。

3-21

解：

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查表，丙烯 $T_c = 365.0K \qquad p_c = 4.620MPa$

则 $T_r = \dfrac{473}{647.3} = 0.7307 \qquad p_r = \dfrac{1.0}{22.05} = 0.0454$

则

$$q = \frac{\Psi}{\Omega} T_r^{-3/2} = \frac{0.42748}{0.08664} \times 0.7307^{-3/2} = 4.3332$$

$$\beta = \Omega \frac{p_r}{T_r} = \frac{0.08664 \times 0.0454}{0.7307} = 0.1720$$

对于水蒸汽，

$$Z = 1 + \beta - q\beta \frac{Z - \beta}{Z(Z + \beta)}$$

利用 $Z = 1$ 的初值进行迭代，收敛结果为 $Z = 0.4291$

则

$$\frac{H^R}{RT} = (Z - 1) - \frac{3}{2} q \ln(\frac{Z + \beta}{Z}) = -2.76178$$

$$H^R = -9138.65 J \cdot mol^{-1}$$

3-24

解：

剩余性质：丙烯 $T_c = 365K \qquad p_c = 4.62Mpa \qquad \omega = 0.148$

1）2.53Mpa & 400K

$$T_r = 1.0959 \qquad p_r = 0.5476$$

$$\frac{\left(H^R\right)^0}{RT_c} = -0.5302 \qquad \frac{\left(H^R\right)^1}{RT_c} = -0.3555$$

$$\frac{\left(H^R\right)}{RT_c} = \frac{\left(H^R\right)^0}{RT_c} + \omega \frac{\left(H^R\right)^1}{RT_c} = -0.5828$$

$$\frac{\left(S^R\right)^0}{R} = -0.3390 \qquad \frac{\left(S^R\right)^1}{R} = -0.3263$$

$$\frac{\left(S^R\right)}{R} = \frac{\left(S^R\right)^0}{R} + \omega \frac{\left(S^R\right)^1}{R} = -0.3873$$

$$Z^0 = 0.8458 \qquad Z^1 = 0.007308$$

$$Z = Z^0 + \omega Z^1 = 0.8469$$

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2） 12.67Mpa & 550K

$T_r = 1.5068 \qquad p_r = 2.7424$

$$\frac{(H^R)^0}{RT_c} = -1.4050 \qquad \frac{(H^R)^1}{RT_c} = 0.0603$$

$$\frac{(H^R)}{RT_c} = \frac{(H^R)^0}{RT_c} + \omega\frac{(H^R)^1}{RT_c} = -1.3961$$

$$\frac{(S^R)^0}{R} = -0.7015 \qquad \frac{(S^R)^1}{R} = -0.1926$$

$$\frac{(S^R)}{R} = \frac{(S^R)^0}{R} + \omega\frac{(S^R)^1}{R} = -0.7300$$

$Z^0 = 0.8034 \qquad Z^1 = 0.22675$

$$Z = Z^0 + \omega Z^1 = 0.8371$$

$$\Delta H_m = \Delta H^{id}{}_m + \Delta H^R{}_m = \int C_p^{id} dT + \Delta H^R{}_m = 11218.16 J \cdot mol^{-1}$$

$$\Delta S_m = \Delta S^{id}{}_m + \Delta S^R{}_m = \int \frac{C_p^{id}}{T} dT - R\int \frac{dp}{p} + \Delta S^R{}_m = 12.81 J \cdot mol^{-1} \cdot K^{-1}$$

$$\Delta V = \Delta(\frac{ZRT}{p}) = -0.00081 m^3 \cdot mol^{-1}$$

3-28

解：

1） $V_1 = 113.640 cm^3 \cdot mol^{-1} \qquad V_2 = 101.460 cm^3 \cdot mol^{-1}$

2）以 0.2 为例

$$\overline{V_1} = V - (1-x_1)\frac{dV}{dx_1} \qquad \overline{V_2} = V - x_1\frac{dV}{dx_1}$$

中心差商

$$\left.\frac{dV}{dx_1}\right|_{x_1=0.2} = \frac{1}{0.2}(-V_{x_1=0.1} + V_{x_1=0.3}) = 12.58$$

则 $\overline{V_1} = 114.086 cm^3 \cdot mol^{-1} \qquad \overline{V_2} = 101.486 cm^3 \cdot mol^{-1}$

3）以 0.2 为例

$$\Delta_{mix} V = V - (x_1 V_1 + x_2 V_2) = 0.106 cm^3 \cdot mol^{-1}$$

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4）

边值差商

$$\left.\frac{dV}{dx_1}\right|_{x_1\to 1}=\frac{1}{0.02}(V_{x_1=1}-V_{x_1=0.98})=11.55$$

$$\overline{V_2^{\infty}}=V_{x_1=1}-\left.\frac{dV}{dx_1}\right|_{x_1\to 1}=102.09\,cm^3\cdot mol^{-1}$$

同理，

$$\overline{V_1^{\infty}}=114.31cm^3\cdot mol^{-1}$$

3-35

解：

$$118^{\circ}F$$

4-8

解：

过程能量守恒，忽略位能和动能的变化，则

$$h_1+w+q=h_2$$

$$W=\frac{w_{s,r}^{s}}{\eta}=\frac{1}{\eta}\left(\frac{k}{k-1}ZRT_1\left(\left(\frac{p_2}{p_1}\right)^{\frac{k-1}{k}}-1\right)\right)\qquad J\cdot mol^{-1}$$

$$Z_1=\frac{p_1V_1}{RT_1}=1.0787\qquad 注M_{CO_2}=44.010g\cdot mol^{-1}$$

$$Z_2=\frac{p_2V_2}{RT_2}=0.8827$$

$$Z=\frac{1}{2}(Z_1+Z_2)=0.9807$$

则

$$w=W/M_{CO_2}\times 1000=3.42\times 10^5 J\cdot kg^{-1}$$

$$q=h_2-h_1-w=-287.75kJ\cdot kg^{-1}$$

放热

4-9

解：

1）7atm

2）~0.13

83-85K

~34kcal/kg

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3）190K

4-21

解：

查水蒸汽表，各点状态如下

1：$(81.33^\circ C, 50\,kPa, H_1 = 340.49\,kJ\cdot kg^{-1})$

泵做功

$w_p = \Delta pV = (1000-50)\times 1.030\times 10^{-3} = 0.9785 kJ\cdot kg^{-1}$　　(忽略体积变化)

2：$(82^\circ C, 1000kPa, H_2 = H_1 + w_p = 351.47 kJ\cdot kg^{-1})$

3：$(500^\circ C, 1000kPa, H_3 = 3478.5 kJ\cdot kg^{-1})$

4：$(81.33^\circ C, 50kPa, H_4 = 2645.9 kJ\cdot kg^{-1})$

则

1）$\eta = \dfrac{H_3 - H_4}{H_3 - H_2} = 26.5\%$

2）$r = \dfrac{\omega_p}{H_3 - H_4} = 0.00118$

3）$m = \dfrac{3600kJ}{H_3 - H_4} = 4.32 kg\cdot h^{-1}$

4-22

解：

各点状态如下：

1：$(-20^\circ C, H_1 = 340.49 kJ\cdot kg^{-1})$

2：$(-20^\circ C, H_2 = 383 kJ\cdot kg^{-1})$

3：$(35^\circ C, H_3 = 423 kJ\cdot kg^{-1})$　　(最高为等熵过程对应值)

4：$(35^\circ C, H_4 = 250 kJ\cdot kg^{-1})$

则

$w = \dfrac{H_2 - H_1}{H_3 - H_2} = 3.325$

$Q = |H_4 - H_3| = 173 kJ\cdot kg^{-1}$

5-10

解：

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丙酮所含元素的标准 yong 为:

$\varepsilon^{\theta}(C)=410.515kJ\cdot mol^{-1}$　　$\varepsilon^{\theta}(H)=117.575kJ\cdot mol^{-1}$　　$\varepsilon^{\theta}(O)=117.575kJ\cdot mol^{-1}$

对于液态，$\Delta_f G^{\theta}(C_3H_6O(l))=-155.73kJ\cdot mol^{-1}$

液态标准 yong $\varepsilon^{\theta}(C_3H_6O(l))=\Delta_f G^{\theta}(C_3H_6O(l))+\alpha\sum\varepsilon_i^{\theta}=1783.242kJ\cdot mol^{-1}$

对于气态，$\Delta_f G^{\theta}(C_3H_6O(g))=-153.05kJ\cdot mol^{-1}$

液态标准 yong $\varepsilon^{\theta}(C_3H_6O(g))=\Delta_f G^{\theta}(C_3H_6O(g))+\alpha\sum\varepsilon_i^{\theta}=1785.922kJ\cdot mol^{-1}$

计算$125^{\circ}C,0.5MPa$下 yong 值，

1.判断汽液态（以对应状态下 yong 值为标准值）　　2.参见公式（5/2-32 5/2-35）

5-11

解：

	参考态			
T	25	250	300	600
p		1.5	5	5
H	104.89	2694.825	2918.5	3666.4
S	0.3674	6.706625	6.205475	7.26825
yong 变	0	699.8951	1072.988	1504.022
质能系数	0	0.270237	0.381356	0.422299

说明：1.以 298.15K 的饱和水为参考态　2.数据由水蒸气表而来（内插法）

3.yong 变　$\Delta\varepsilon=(H-H_0)+T_0(S-S_0)$

4.质能系数 $\lambda=\Delta\varepsilon/(H-H_0)$

5.温度越高，压力越大，质能系数越大

5-14

解：

参见 5/3-2，注意内部 yong 损失

6.8

解：

临界性质：Tc=425.2K　　Pc=3.799Mpa　　w=0.193

对比性质：Tr=1.0818　　Pr=0.3948

对于 RK 方程

$$q=\frac{0.42748}{0.08664}Tr^{-3/2}=4.3848$$

$$\beta=0.08664\frac{\mathrm{Pr}}{Tr}=0.03162$$

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$$Z_{k+1}=1+\beta-q\beta\frac{Z_K-\beta}{Z_k(Z_k+\beta)}$$

以 Z_0=1 代入上式迭代，Z=0.9131

$$I=\ln\frac{Z+\beta}{Z}=0.03404$$

$$\ln\phi=Z-1-\ln(Z-\beta)-qI=-0.1100$$

$$\phi=0.8958$$

$$f=\phi P=1.344MPa$$

6.9

解：

临界性质：Tc=365.0K　　Pc=4.620Mpa　　w=0.148

对比性质：Tr=1.3096　　Pr=1.4892

查附录 B5 内差可得

$$\phi^0=0.79992\qquad \phi^1=1.119381$$

$$\phi=\phi^0(\phi^1)^w=0.8134$$

$$f=\phi P=5.596MPa$$

6.10

解：

以 200℃，6kPa 的水蒸气状态作为理想气体，$f_i^*=p_i^*=6kPa$

则

$$h_i^*=2879.7J\cdot g^{-1}\qquad s_i^*=9.1398J\cdot g^{-1}\cdot K^{-1}$$

$p=1Mpa$，

$$h_i=2827.9J\cdot g^{-1}\qquad s_i=6.6940J\cdot g^{-1}\cdot K^{-1}$$

$$\ln\frac{f_i}{f_i^*}=\frac{M(H_2O)}{R}\left[\frac{h_i-h_i^*}{T}-(s_i-s_i^*)\right]=5.0624$$

$$\frac{f_i}{f_i^*}=157.6967\qquad f_i=947.82kPa$$

$$\phi_i=\frac{f_i}{p}=0.9478$$

6.11

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解：

临界性质：Tc=369.2K　　Pc=4.975MPa

计算饱和蒸汽的逸度系数（Ps=0.267Mpa）

对比性质：Tr=0.82611　　Pr=0.05367

对于 RK 方程

$$q=\frac{0.42748}{0.08664}Tr^{-3/2}=6.5711$$

$$\beta=0.08664\frac{\Pr}{Tr}=0.005629$$

$$Z_{k+1}=1+\beta-q\beta\frac{Z_K-\beta}{Z_k(Z_k+\beta)}$$

以 Z_0=1 代入上式迭代，Z=0.9702

$$I=\ln\frac{Z+\beta}{Z}=0.005785$$

$$\ln\phi^s=Z-1-\ln(Z-\beta)-qI=-0.03174$$

$$\phi^s=0.9688$$

$$V^l=4.8\times10^{-4}\times86.469\times10^{-3}=4.1505\times10^{-5}m^3\cdot kg^{-1}$$

$$f^l=\phi^sP^s\exp[\frac{V^l}{RT}(P-P^s)]=0.3238MPa$$

6-15

解：

1.溶解度参数法

查 B10 表，可得

环己烷 $\delta_1=16.77(MJ\cdot m^{-3})^{1/2}$　　$V_1^l=0.1088m^3\cdot kmol^{-1}$

苯　　$\delta_2=18.23(MJ\cdot m^{-3})^{1/2}$　　$V_2^l=0.0894m^3\cdot kmol^{-1}$

又有 $x_1=0.4$　　$x_2=0.6$　　$T=313.15K$

则

$$\ln\gamma_1=\frac{V_1^lx_2^2}{RT}(\delta_1-\delta_2)^2=0.058385$$

$$\ln\gamma_2=\frac{V_2^lx_1^2}{RT}(\delta_1-\delta_2)^2=0.021322$$

所以

$$\gamma_1=1.0601\quad a_1=\gamma_1x_1=0.4240$$

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$$\gamma_2 = 1.0216 \quad a_2 = \gamma_2 x_2 = 0.6129$$

2.Margules 模型

知 $A_{12} = 0.1640 \quad A_{21} = 0.1462$

则

$$\ln\gamma_1 = x_2^2\left[A_{12} + 2\left(A_{21} - A_{12}\right)x_1\right] = 0.0539136$$

$$\ln\gamma_2 = x_1^2\left[A_{21} + 2\left(A_{12} - A_{21}\right)x_2\right] = 0.0268096$$

所以

$$\gamma_1 = 1.0554 \quad a_1 = \gamma_1 x_1 = 0.4222$$

$$\gamma_2 = 1.0272 \quad a_2 = \gamma_2 x_2 = 0.6163$$

6-17

解：

气相视为理想气体，则有

$$y_i p = x_i \gamma_i p_i^s$$

另

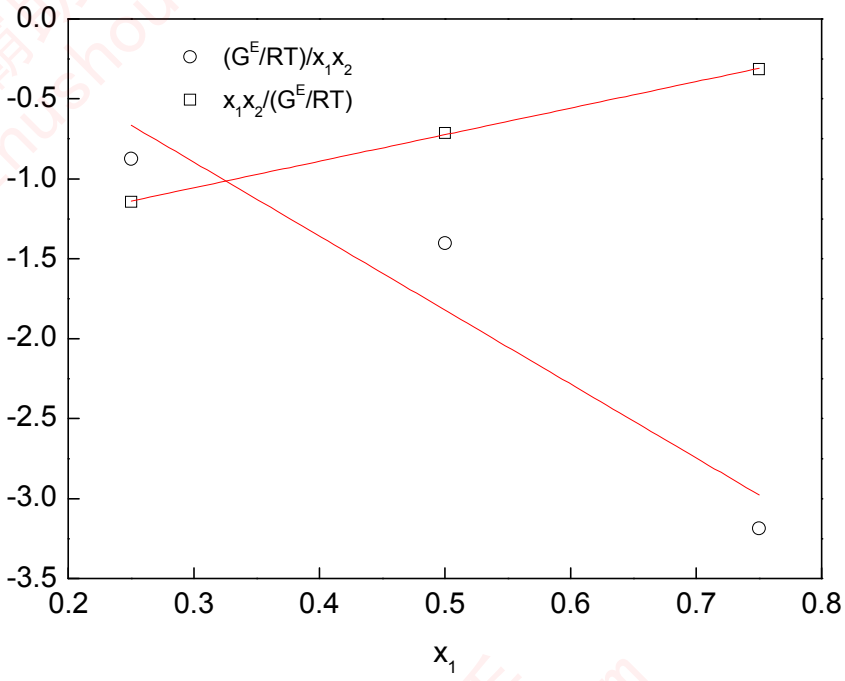
$$\frac{G^E}{RT} = \sum x_i \gamma_i$$

故

p/Mpa	x_1	y_1	$\ln\gamma_1$	$\ln\gamma_2$	$\frac{G^E}{RT}$	$\frac{G^E/RT}{x_1x_2}$	$\frac{x_1x_2}{G^E/RT}$
0.3018	0.25	0.188	-0.68005	0.00814	-0.16391	-0.87417	-1.14394
0.2768	0.50	0.378	-0.76121	0.060575	-0.35032	-1.40127	-0.71364
0.2487	0.75	0.545	-0.90783	0.334032	-0.59737	-3.18596	-0.31388

作图如下：

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所以 $\frac{x_1x_2}{G^E/RT}$ 与 x_1 线性关系较好，选用 van Laar 模型。

6-18

解：

由 $G^E/RT = x_1x_2[A+b(1-2x_1)]$ 和 $G^E/RT = x_1x_2(A_{21}x_1 + A_{12}x_2)$，则

$$A_{21} = A - B \quad A_{12} = A + B$$

又有

$$\ln\gamma_1 = x_2^2[A_{12} + 2(A_{21} - A_{12})x_1]$$

$$\ln\gamma_2 = x_1^2[A_{21} + 2(A_{12} - A_{21})x_2]$$

$$\frac{G^E}{RT} = \sum x_i\gamma_i$$

由上述公式，可得

T/K	x_1	A_{12}	A_{21}	$\ln\gamma_1$	$\ln\gamma_2$	$\frac{G^E}{RT}$
109	0.5	0.2867	0.3205	0.080125	0.071675	0.0759
112	0.5	0.3062	0.2826	0.07065	0.07655	0.0736
115.74	0.5	0.335	0.2258	0.05645	0.08375	0.0701

1.活度系数

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$\gamma_1 = 1.0732 \quad \gamma_1 = 1.0796$

2.混合热

$$\Delta_{mix}H = H^E = -RT^2\left(\frac{\partial\left(G^E/RT\right)}{\partial T}\right) = -8.314\times112^2\times\left(\frac{0.0701-0.0759}{115.74-109}\right) = 89.74\text{J}\cdot\text{mol}^{-1}$$

3.超额熵

$$S^E = \frac{H^E - G^E}{T} = \frac{89.74 - 0.0736\times8.314\times112}{112} = 0.189\text{J}\cdot\text{mol}^{-1}\cdot K^{-1}$$

7-19

解：

1）$353.15K$ 时有

$$V_1^l = 76.92cm^3\cdot mol^{-1} \qquad V_2^l = 18.07cm^3\cdot mol^{-1}$$

$$\ln p_1^s = 16.6780 - \frac{3640.20}{T-53.54} = \qquad p_1^s = 92.59kPa$$

$$p_2^s = 47.39kPa(\text{水蒸汽表})$$

$a_{12} = 4250.11J\cdot mol^{-1}$ $a_{21} = 5374.81J\cdot mol^{-1}$，则

$$\Lambda_{ij} = \frac{V_j}{V_i}\exp\frac{-a_{ij}}{RT}(i\neq j)$$

$$\Lambda_{12} = 0.055241 \qquad \Lambda_{21} = 0.682434$$

$x_1 = 0.25$，则

$$\ln\gamma_1 = -\ln(x_1 + x_2\Lambda_{12}) + x_2\left(\frac{\Lambda_{12}}{x_1+x_2\Lambda_{12}} - \frac{\Lambda_{21}}{x_2+x_1\Lambda_{21}}\right) = 0.819152$$

$$\ln\gamma_2 = -\ln(x_2 + x_2\Lambda_{21}) - x_1\left(\frac{\Lambda_{12}}{x_1+x_2\Lambda_{12}} - \frac{\Lambda_{21}}{x_2+x_1\Lambda_{21}}\right) = 0.220654$$

$$\gamma_1 = 2.268575 \qquad \gamma_2 = 1.246892$$

气相视为理想气体，则有

$$y_i p = x_i\gamma_i p_i^s$$

$$\begin{matrix} y_1 p = 0.25\times2.268575\times92.59233 \\ y_2 p = (1-0.25)\times1.246892\times47.39 \end{matrix}$$，且 $y_1 + y_2 = 1$，则

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$y_1 = 0.5423 \quad y_2 = 0.4577 \quad p = 96.83kPa$

2) 已知 $y_1 = 0.4 \quad y_2 = 0.6 \quad p = 101.325$

假设迭代初始温度为 $T0 = 353.15$，则

$$V_1^l = 76.92cm^3 \cdot mol^{-1} \qquad V_2^l = 18.07cm^3 \cdot mol^{-1}$$

$$\ln p_1^s = 16.6780 - \frac{3640.20}{T - 53.54} = \qquad p_1^s = 92.59kPa$$

$p_2^s = 47.39kPa$(水蒸汽表）

假设迭代初始 $\gamma_1 0 = \gamma_2 0 = 1.0$，则

$$y_i p = x_i \gamma_i p_i^s$$

$$x_1 = 0.437726 \quad x_2 = 1 - x_1 = 0.562274$$

计算 γ_1 和 γ_2

$a_{12} = 4250.11J \cdot mol^{-1} \quad a_{21} = 5374.81J \cdot mol^{-1}$，则

$$\Lambda_{ij} = \frac{V_j}{V_i} \exp \frac{-a_{ij}}{RT} (i \neq j)$$

$$\Lambda_{12} = 0.055241 \quad \Lambda_{21} = 0.682434$$

$$\ln \gamma_1 = -\ln(x_1 + x_2\Lambda_{12}) + x_2\left(\frac{\Lambda_{12}}{x_1 + x_2\Lambda_{12}} - \frac{\Lambda_{21}}{x_2 + x_1\Lambda_{21}}\right) = 0.3782$$

$$\ln \gamma_2 = -\ln(x_2 + x_2\Lambda_{21}) - x_1\left(\frac{\Lambda_{12}}{x_1 + x_2\Lambda_{12}} - \frac{\Lambda_{21}}{x_2 + x_1\Lambda_{21}}\right) = 0.445035$$

$$\gamma_1 = 1.459655 \quad \gamma_2 = 1.560545$$

计算新的 p_1^s

$$p_1^s = p\left(\frac{y_1}{\gamma_1} + \frac{y_2}{\gamma_2}\frac{p_1^s}{p_2^s}\right) = 103.8835$$

计算新的 T

$$T = \frac{-3640.20}{\ln p_1^s - 16.6780} + 53.54 = 356.0146K$$

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以新的 T 问初始温度进行上述迭代，注意水蒸气表的利用内插取值，每一步迭代中要用的γ_i和p_i^s为上次迭代的结果，直至收敛。

$x_1=0.0296 \quad x_2=0.9704$

$\gamma_1=11.3375 \quad \gamma_2=1.0100$

$p_1^s=120.80\,kPa \quad p_2^s=62.03kPa$

$T=359.85K$

7-16

解：

气相视为理想气体，则有

$$y_i p=x_i\gamma_i p_i^s$$

另

$$\frac{G^E}{RT}=\sum x_i\ln\gamma_i$$

1）$\gamma_1=\dfrac{y_1p}{x_1p_1^s}=\dfrac{0.634\times24.4}{0.300\times23.1}=2.2323$

$\gamma_2=\dfrac{y_2p}{x_2p_2^s}=\dfrac{(1-0.634)\times24.4}{(1-0.300)\times10.05}=1.2694$

2）$\dfrac{G^E}{RT}=\sum x_i\ln\gamma_i=0.300\times\ln2.2323+(1-0.300)\times\ln1.2694=0.4079$

3）$\dfrac{\Delta_{mix}G}{RT}=\dfrac{G^E}{RT}+\dfrac{\Delta_{mix}G^{id}}{RT}=\dfrac{G^E}{RT}+\dfrac{\sum x_i\ln x_i}{RT}=-0.2030$

4）活度系数大于 1，故正偏差

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